

Task 1: Create Two 200 MB Partitions

Check available disks:

- `lsblk`

Suppose your new disk is `/dev/sdb`.

Open `fdisk`:

- `sudo fdisk /dev/sda`

Inside `fdisk`, create two partitions:

- `n`
- `p`
- `1`
- `+200M`
- `n`
- `p`
- `2`
- `+200M`
- `w`

Verify:

- `lsblk`

Task 2: Create Physical Volumes (PV)

Initialize both partitions:

- `sudo pvcreate /dev/sdb1 /dev/sdb2`

Verify:

- `pvs`

Task 3: Create Volume Group (VG)

Create the volume group named `musicvg`:

- `sudo vgcreate musicvg /dev/sdb1 /dev/sdb2`

Verify:

- `vgdisplay musicvg`

Task 4: Create Logical Volume

Create a 300 MB logical volume named `playlistlv`:

- `sudo lvcreate -L 300M -n playlistlv musicvg`

Verify:

- `lvs`

Format it:

- `sudo mkfs.ext4 /dev/musicvg/playlistlv`

Create the mount point:

- `sudo mkdir -p /mnt/playlist`

Mount it:

- `sudo mount /dev/musicvg/playlistlv /mnt/playlist`

Verify:

- `df -h`

Task 5: Extend the Logical Volume by 50 MB

Extend the logical volume:

- `sudo lvextend -L +50M /dev/musicvg/playlistlv`

Resize the ext4 filesystem:

- `sudo resize2fs /dev/musicvg/playlistlv`

Verify the updated size:

- `df -h /mnt/playlist`

Check LVM details:

- `lvs`